

How robust are traditional stock price models under bubble-like market dynamics?

Analysis of the AI bubble

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Abstract

This paper examines the classical Black-Scholes option pricing framework, which assumes that asset prices follow a geometric Brownian motion, and applies it to stock price dynamics of AI-related firms. While the model provides a tractable and widely used benchmark for option pricing, its underlying assumptions—particularly constant volatility and log-normal returns—are often too restrictive to capture empirical market behavior.

In particular, we argue that the Black-Scholes framework is limited in its ability to reflect periods of rapidly changing volatility and bubble-like price dynamics observed in high-growth sectors such as artificial intelligence. To address these shortcomings, we introduce more flexible time-series approaches based on ARMA-GARCH models, which allow for conditional heteroskedasticity and more realistic volatility clustering.

Keywords: Black-Scholes model, artificial intelligence bubble, ARMA-GARCH models, volatility modeling, simulation.

Introduction

The rapid development of artificial intelligence has had a profound impact not only on daily life and future labor markets, but also on global financial markets and large technological enterprises. Artificial intelligence has evolved beyond a simple computational tool and has become increasingly integrated into research, scientific experimentation, optimization, and large-scale decision-making processes used by both corporations and governments.

As a result, investment in artificial intelligence technologies and infrastructure has grown dramatically over the past decade. In particular, the extraordinary increase in demand for AI services, cloud computing, and semiconductor technology has led to substantial growth in the market valuations of AI-focused firms. This rapid appreciation in stock prices has raised concerns regarding the possible formation of speculative bubbles within AI-related assets.

Major technology companies such as NVIDIA, Meta Platforms, and Microsoft have experienced exceptional increases in their stock prices during the post-pandemic period, particularly following the emergence of large-scale generative AI systems.

This naturally raises the following question: to what extent can traditional financial models remain relevant in describing or predicting the dynamics of AI-related equities during speculative market periods?

Among the most influential classical financial models is the Black–Scholes framework, developed in 1973, which introduced one of the most widely used option pricing formulas in quantitative finance. The model assumes that stock prices follow a geometric Brownian motion with constant volatility and continuous dynamics.

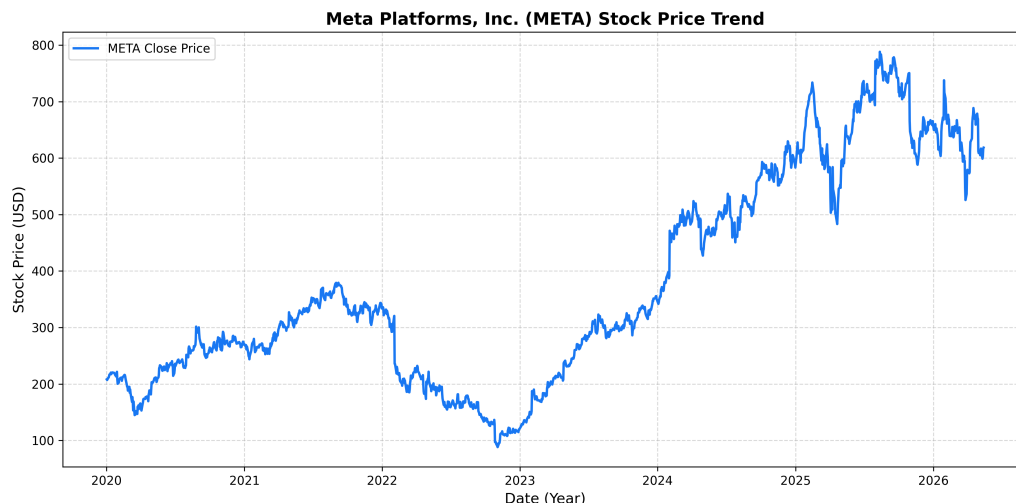


Figure 1: META closing stock price since 2020.

As illustrated in Figure 1, META experienced substantial volatility during the post-pandemic AI expansion period. Such behavior raises important questions regarding the suitability of traditional diffusion-based models under highly speculative market conditions characterized by volatility clustering, rapid price appreciation, and abrupt market corrections.

This paper investigates whether classical stochastic models such as Black–Scholes remain informative in the context of AI-related equities, and explores the extent to which more advanced models incorporating stochastic volatility or jump dynamics may better capture the statistical behavior of speculative assets.

Black-Scholes Model

Black-Scholes Formula for Option Pricing

In 1973, Fisher Black and Myron Scholes published their groundbreaking work option pricing. Indeed, in their article "The Pricing of Options and Corporate Liabilities" [BS73], they derived the Black-Scholes formula for option pricing:

$$C^{BS}(t, S; r, \sigma, K, T) := S\Phi(d_1) - Ke^{-r(T-t)}\Phi(d_2) \quad (1)$$

where Φ denotes the standard normal distribution (df), r represents the continuously compounded risk-free interest rate, σ denotes the volatility of the underlying stock, and where

$$d_1 = \frac{\ln \frac{S}{K} + (r + \frac{1}{2}\sigma^2)(T-t)}{\sigma\sqrt{T-t}} \quad d_2 = d_1 - \sigma\sqrt{T-t} \quad (2)$$

Here, K is the premium, while T is the maturity of the option.

We will use standard market practices where time is measured in years and r and σ are expressed in annualized terms.

Furthermore, we will set S_t to be the price of the stock at time t . Real market interest rates and volatilities change constantly and thus depend on time: r_t, σ_t . As a result, we take the vector of risk-factors as a function of time: $\mathbf{Z}_t = (\ln S_t, r_t, \sigma_t)'$.

The risk-factor changes are given by :

$$\mathbf{X}_{t+1} = (\ln S_{t+1} - \ln S_t, r_{t+1} - r_t, \sigma_{t+1} - \sigma_t)'. \quad (3)$$

Geometric Brownian Motion for Stock Prices

We will use the [Shr04] : First consider a portfolio where we invest in a money market account paying a constant rate of interest r and in a stock modeled by the geometric Brownian motion

$$dS(t) = \alpha S(t)dt + \sigma S(t)dW(t) \quad (4)$$

Suppose at each time t , the investor holds $\Delta(t)$ shares of stock. The remainder of the portfolio value, $X(t) - \Delta(t)S(t)$, is invested in the money market account. Thus the differential $dX(t)$ for the investor's portfolio value at time t is due to the following 2 factors : capital gain $\Delta(t)dS(t)$ on the stock position and the interest earnings $r(X(t) - \Delta(t)S(t))dt$ on the cash position, and thus amounts to :

$$dX(t) = \Delta(t)dS(t) + r(X(t) - \Delta(t)S(t))dt = \Delta(t)(\alpha S(t)dt + \sigma S(t)dW(t)) + r(X(t) - \Delta(t)(\alpha S(t)dt + \sigma S(t)dW(t)))dt = rX(t)dt +$$

As explained in [Shr04], the three terms can be understood as :

1. $rX(t)dt$ an average underlying rate of return r on the portfolio
2. $\Delta(t)(\alpha - r)S(t)dt$ a risk premium $\alpha - r$ for investing in the stock
3. $\Delta(t)\sigma S(t)dW(t)$ a volatility term proportional to the size of the stock investment

We seek a continuous function $c(t, x)$ that is a solution to the Black-Scholes-Merton partial differential equation

$$c_t(t, x) + rx c_x(t, x) + \frac{1}{2}\sigma^2 x^2 c_{xx}(t, x) = rc(t, x), \quad \forall t \in [0, T], \quad x \geq 0 \quad (5)$$

And that satisfies the terminal condition

$$c(T, x) = (x - K)^+ \quad (6)$$

The solution to 5 is the following :

$$c(t, x) = x\Phi(d_1) - Ke^{-r(T-t)}\Phi(d_2), \quad 0 \leq t < T, \quad x > 0 \quad (7)$$

with d_1, d_2 as described above 2.

Although the Black-Scholes framework is primarily known for its option pricing formula, it is built on the assumption that the underlying stock price follows a geometric Brownian motion,

$$dS(t) = \alpha S(t)dt + \sigma S(t)dW(t).$$

In the Example 6.2.1 on [Shr04], we derive the solution to this stochastic differential equation and find that when we set initial time zero and initial position to be $S(0)$, namely :

$$S(t) = S(0) \exp \left(\left(\alpha - \frac{1}{2}\sigma^2 \right)t + \sigma W(t) \right)$$

Similarly, for $T \geq t$: $S(T) = S(0) \exp \left(\left(\alpha - \frac{1}{2}\sigma^2 \right)t + \sigma W(t) \right)$

However, the Brownian increment property states the following : $W_{t+\Delta t} - W_t \sim \sqrt{\Delta t}Z_t \quad Z_t \sim N(0, 1)$

Therefore, for simulation over a discrete time step Δt , we use

$$S_{t+\Delta t} = S_t \exp \left(\left(\alpha - \frac{1}{2}\sigma^2 \right)\Delta t + \sigma\sqrt{\Delta t}Z_t \right), \quad Z_t \sim N(0, 1).$$

This is the formula used to generate the simulated Black-Scholes stock price path.

Simulation of META Stock Price using the Geometric Brownian Motion derived from the Black-Scholes Formula

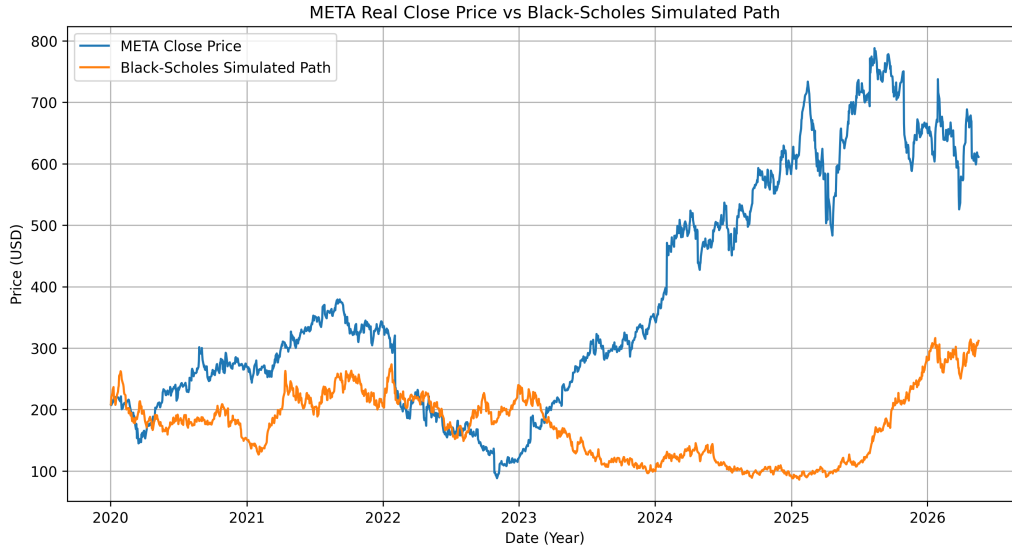


Figure 2: META closing stock price vs Black Scholes Simulated path since 2020.

New Methods

Times Series Analysis : ARMA-GARCH Model

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AI Stress-Testing

Conclusion

References

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